DeVinci Blockchain

# Git & GitHub Cheat sheet

DVB WORKSHOP
22.10.2026 · IIM NANTERRE

KEEP IT OPEN WHILE YOU CODE

BLUE = REPLACE WITH YOUR OWN VALUE

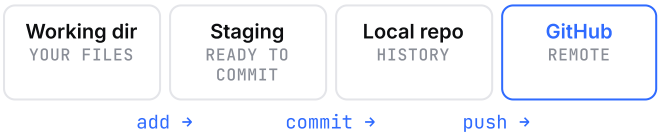
## 01 Set up (once per computer)

| | |
|---|---|
| `git config --global user.name "First Last"` | Your name, shown on every commit. |
| `git config --global user.email "you@mail.com"` | The **same e-mail** as your GitHub account. |
| `git config --global init.defaultBranch main` | New repos start on the **main** branch. |
| `git config --global core.editor "code --wait"` | Opens VS Code instead of Vim for messages. |
| `git config --list` | Check your configuration. |

## 02 Start a project

| | |
|---|---|
| `git init` | Turns the current folder into a Git repo. |
| `git clone <url>` | Copies a remote repo to your machine. |
| `git --version` | Checks that Git is installed. |

## 03 The basic cycle




| | |
|---|---|
| `git status` | Where am I? **Type it all the time.** |
| `git add file` | Adds a file to the staging area. |
| `git add .` | Adds every change in the folder. |
| `git commit -m "message"` | Saves a "snapshot" of the staging area. |
| `git diff` | Changes not added yet. |
| `git diff --staged` | Changes ready to be committed. |
| `git log --oneline --graph --all` | The history as a compact graph. |

## 04 Branches

| | |
|---|---|
| `git branch` | Lists branches (* = current branch). |
| `git switch -c feature/my-card` | Creates a branch and moves onto it. |
| `git switch main` | Goes back to main. |
| `git merge feature/my-card` | Merges that branch **into the current one**. |
| `git branch -d feature/my-card` | Deletes a branch that is already merged. |

## 05 Working with GitHub (remote)

| | |
|---|---|
| `git remote -v` | Shows the remote repos (origin...). |
| `git remote add origin <url>` | Links your local repo to GitHub. |
| `git push -u origin branch` | First push of a branch (then just `git push`). |
| `git pull` | Downloads **and merges** new changes. |
| `git fetch` | Downloads without merging anything. |

## 06 Undo

| | |
|---|---|
| `git restore file` | Discards changes **not added yet** (gone for good!). |
| `git restore --staged file` | Unstages a file, keeps the changes. |
| `git commit --amend -m "msg"` | Fixes the last commit (before pushing). |
| `git revert <hash>` | New commit that undoes an old one. **Safe** even after a push. |
| `git reset --soft HEAD~1` | Undoes the last commit, keeps the changes staged. |

⚠ **Careful:** `amend` and `reset` rewrite history. Never use them on a commit that is already pushed and shared: use `revert`.

**The Builders Wall**
Add your card with a Pull Request:
`github.com/alibenyezza/dvb-git-workshop`

## 07 Fix a merge conflict

Git can't merge two changes to the same line, so it shows you both versions:

```
<<<<<<< HEAD
Welcome to the Git workshop!   ← your version
=======
Welcome to the DVB workshop!   ← the other branch
>>>>>>> feature/title
```

1. **Open the file** (VS Code highlights the conflict).
2. **Keep the right version** (or combine them) and **delete the 3 markers**.
3. `git add file` to mark the conflict as resolved.
4. `git commit` to finish the merge.

| | |
|---|---|
| `git merge --abort` | Panicking? Cancels the merge and goes back to how things were. |

## 08 Contribute with a Pull Request

`Fork` → `Clone` → `Branch` → `Commit` → `Push` → `PR` → `Review` → `Merge`

| | |
|---|---|
| `"Fork" button on GitHub` | Creates your own copy of the repo on your account. |
| `git clone https://github.com/you/dvb-git-workshop.git` | Clone **your fork**, not the original. |
| `git switch -c add/username` | One branch for your contribution. |
| `builders/username.json` | Create your card (copy `_template.json`). |
| `git add builders/username.json`<br>`git commit -m "feat: add username to the wall"` | Saves your card. |
| `git push -u origin add/username` | Sends the branch to your fork. |
| `"Compare & pull request"` | On GitHub. Wait for the **validate** ✓ check, then the merge. |

## 09 Write good commits

**Conventional Commits** format: `type: short description`

| | |
|---|---|
| `feat:` | new feature — `feat: add login page` |
| `fix:` | bug fix — `fix: broken login link` |
| `docs:` | documentation — `docs: update README` |
| `chore:` | maintenance — `chore: bump dependencies` |
| `refactor:` | rewrite without changing behaviour |

Short (< 72 characters), imperative mood, **one commit = one change**.

## 10 Versions & tags

**SemVer**: `MAJOR.MINOR.PATCH` — breaking change · new feature · bug fix.

| | |
|---|---|
| `git tag -a v1.0.0 -m "First release"` | Creates an annotated tag on the current commit. |
| `git tag` | Lists tags. |
| `git push origin v1.0.0` | Pushes a tag (a plain `git push` doesn't send tags). |
| `git push --tags` | Pushes every tag. |

## 11 Help!

| | |
|---|---|
| `Stuck in Vim` | **Esc** then `:wq` + Enter (save) or `:q!` (quit without saving). |
| `Author identity unknown` | Set `user.name` and `user.email` (section 01). |
| `! [rejected] … (fetch first)` | The remote moved on: `git pull --rebase` then `git push`. |
| `You are in 'detached HEAD'` | Go back to a branch: `git switch -` or `git switch main`. |
| `not a git repository` | Wrong folder: `cd` into your repo (or `git init`). |
| `Authentication failed` | GitHub no longer accepts passwords: sign in through the browser on your 1st push, or run `gh auth login`. |
| `Permission denied (403)` | You're pushing to the original repo: push to **your fork**. |

**Keep learning**

Official docs: `git-scm.com/book` · Practice: `learngitbranching.js.org`
Student Developer Pack: `education.github.com/pack`

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# Keep learning

VIDEOS · GAMES · DOCS
OPEN SOURCE · CERTIFICATIONS

EVERY TITLE IS A CLICKABLE LINK IN THE PDF

FREE = COSTS NOTHING

## 12 Watch

- **Git Explained in 100 Seconds** ↗ 2 MIN
  The big picture, fast · *Fireship*
- **How Git Works: Explained in 4 Minutes** ↗ 4 MIN
  The 4 areas, visually · *ByteByteGo*
- **Git and GitHub for Beginners, Crash Course** ↗ BEGINNER
  Everything we did at the workshop, step by step · *freeCodeCamp*
- **Learn Git in 1 Hour** ↗ BEGINNER
  Very clear, great to start · *Programming with Mosh*
- **Git & GitHub Crash Course** ↗ BEGINNER
  *Traversy Media*
- **Git and GitHub Tutorial for Beginners** ↗ VS CODE
  With VS Code and GitHub Desktop · *Kevin Stratvert*
- **Version Control (git)** ↗ HOW IT WORKS
  How Git works inside: the MIT lecture · *Missing Semester*
- **Git for Professionals** ↗ NEXT LEVEL
  Rebase, cherry-pick, reflog, branching strategies · *freeCodeCamp*
- **13 Advanced (but useful) Git Techniques** ↗ TIPS
  *Fireship*

## 13 Play & visualise FREE

- **Learn Git Branching** ↗
  Levels to master branches, merge and rebase, in your browser
- **Oh My Git!** ↗
  A card game that teaches Git
- **Visualizing Git** ↗
  Type commands and watch the commit graph change

## 14 Learn by doing on GitHub FREE

- **GitHub Skills** ↗
  Courses inside a repo: Intro to GitHub ↗, Review PRs ↗, Resolve conflicts ↗
- **Microsoft Learn: GitHub Foundations** ↗
  The official path, prepares the GH-900 exam
- **First Contributions** ↗
  Your 1st open-source PR, guided (like the Builders Wall!)
- **Good First Issue** ↗
  Beginner-friendly issues in real open-source projects

## 15 Read

- **Pro Git** ↗ FREE
  The official Git book, chapters 1–3 are a must
- **GitHub Docs: Hello World** ↗
  Then the whole GitHub Docs ↗
- **Atlassian Git tutorials** ↗
  Clear articles on merge vs rebase and team workflows
- **Missing Semester: version control** ↗
  MIT notes + exercises
- **Oh Shit, Git!?!** ↗ · **Git Flight Rules** ↗
  What to do when things go wrong
- **GitHub's official Git cheat sheet** ↗
  Another one-page reminder

## 16 Tools & certification

- **VS Code Source Control** ↗ · **GitHub Desktop** ↗ · **GitHub CLI** ↗
  Git with buttons, or `gh` for PRs from the terminal
- **GitHub Foundations certification** ↗ FREE · STUDENTS
  GH-900: claim the voucher in the Student Developer Pack ↗
- **Next: CI/CD & GitHub Actions**
  See our second cheat sheet: `resources/cicd-cheatsheet.pdf`

**All the resources of the workshop**
Slides, cheat sheets and docs for every part:
`github.com/alibenyezza/dvb-git-workshop/tree/main/resources`

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